I am ready for Midterm 2 if

- I remember the definitions of:
  - ☐ discrete random variable
  - ☐ probability mass function (p.m.f.)
  - ☐ cumulative distribution function (c.m.f.)
  - ☐ continuous random variable
  - ☐ probability density function (p.m.f.)
  - ☐ expectation, moments, variance, covariance
  - ☐ probability generating function (p.g.f.) and moment generating function (m.g.f.) and their properties
- I know the following distributions (p.m.f./p.d.f.) and their basic properties:
  - ☐ binomial $\mathscr{B}(n, p)$
  - ☐ Poisson $\mathscr{B}(\lambda)$
  - ☐ geometric $\mathscr{G}(p)$
  - ☐ uniform $\mathscr{U}(a, b)$
  - ☐ normal $\mathscr{N}(\mu, \sigma^2)$
  - ☐ exponential $\mathscr{E}(\lambda)$
- I understand the following notions:
  - ☐ joint distribution of r.v.'s (joint p.m.f./p.d.f., joint c.d.f.)
  - ☐ marginal distribution
  - ☐ independence of r.v.'s
  - ☐ covariance and correlation
  - ☐ conditional distribution of X given that $Y = y$ (conditional p.m.f./p.d.f.)
  - ☐ distribution of transformed r.v.'s (or random vectors)
- I can:
  - ☐ recognise standard discrete and continuous distributions
  - ☐ construct random variables corresponding to given experiments
  - ☐ calculate distributions of random variables
  - ☐ calculate expectation, variance, and moments of a r.v.
  - ☐ calculate joint and marginal distributions of r.v.'s and their covariance
  - ☐ determine whether r.v.'s are independent
  - ☐ determine the conditional distribution of a r.v. given the value of the other r.v.
  - ☐ determine distribution of transformations of r.v.'s (e.g. of the sum $X + Y$)
  - ☐ solve standard problems about r.v.'s and their distributions (e.g. from the problem sets 4–6)

Yes, I am now ready for Midterm 2!